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Latent Memory Palace

Autoregressive Variational Inference Brings Chain-of-Thought Deliberation to Physical Control

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Language models reason flexibly by generating variable-length chains of thought before producing an answer. Continuous control policies cannot: they map observations directly to actions in a fixed-depth computation graph, with no mechanism to think longer on harder problems. Latent Memory Palace (LMP) closes this gap by formalizing deliberation as autoregressive variational inference over a structured latent state space, enabling physical agents to allocate more or fewer computation steps depending on task complexity—achieving 16–26% higher success rates than fixed-depth recurrent baselines on long-horizon manipulation tasks.

AT A GLANCE

Problem: Continuous control policies cannot adaptively allocate deliberation time; they use the same computation depth on easy and hard decisions.

Why it matters: Long-horizon manipulation and dexterous tasks require proportionally deeper deliberation than routine motions, yet existing methods offer no mechanism for this.

Method: Hierarchical variational autoencoder with autoregressive latent steps and a learned halting mechanism trained via ELBO maximization.

Result: 47% higher success rate on long-horizon manipulation versus best recurrent baseline (LSTM); 16–26% across all tasks.

Evidence: Five-task benchmark spanning stacking, tool use, pick-place, deformable manipulation, and multi-object coordination.

The Big Picture

The gap between language model reasoning and physical control is not merely architectural — it is computational. A language model generating a plan for a complex task can produce a chain of thought of any length: simple tasks get short chains, complex tasks get long ones. This adaptive computation is free in the discrete-token genera-

tion paradigm where each token takes roughly the same wall-clock time.

Continuous control policies lack this flexibility entirely. A recurrent policy (LSTM, GRU, S4) updates a fixed-size hidden state once per time step. A diffusion policy iterates a fixed number of denoising steps. Neither can elect to “think harder” about a particularly tricky configuration. The result is policies that perform well on routine subtasks but fail catastrophically when unexpected complexity arises.

Latent Memory Palace formalizes variable-depth deliberation in the continuous domain by casting it as inference in a hierarchical latent variable model where the number of latent computation steps is itself a learned variable.

Why Existing Approaches Fall Short

Recurrent policies (LSTM, GRU, S4) maintain a hidden state that accumulates information over time but applies the same fixed-depth computation at each action step. The hidden state is informative but the per-step computation is shallow.

Model-based RL (Dreamer, TDMPC) plans over fixed-depth horizon rollouts. The horizon must be pre-specified and cannot adapt to task difficulty without expensive online search.

Diffusion policies (Diffuser) perform iterative denoising in action space. This iteration serves trajectory optimization, not task-level deliberation — the number of denoising steps is fixed and does not correlate with decision complexity.

Universal Transformers adapt computation depth in sequence modeling but have not been extended to continuous RL or equipped with the variational training objective needed for learning under environment uncertainty.

Core Mechanism

LMP introduces a *latent deliberation sequence*: before producing each action a_t , the agent runs K_t autoregressive latent computation steps $z_t^{(1)}, \dots, z_t^{(K_t)}$. Each latent step refines the agent’s internal representation of the task state, analogous to one token of chain-of-thought reasoning. The final latent state $z_t^{(K_t)}$ is then decoded to an action.

The number of steps K_t is itself learned: a halt probability $p_h(z_t^{(k)}, o_t) \in [0, 1]$ is computed at each step, and the agent halts when a Bernoulli sample fires. Harder observations (estimated via an auxiliary complexity measure) produce lower halt probabilities, naturally inducing more deliberation.

Technical Formulation

The model defines a hierarchical generative process with autoregressive latent prior:

$$p_{\theta}(z_t^{(1:K)} | o_t) = \prod_{k=1}^K p_{\theta}(z_t^{(k)} | z_t^{(1:k-1)}, o_t)$$

and posterior using retrospective information h_t^{ret} (future observations and rewards, available during training):

$$q_{\phi}(z_t^{(1:K)} | o_t, h_t^{\text{ret}}) = \prod_{k=1}^K q_{\phi}(z_t^{(k)} | z_t^{(1:k-1)}, o_t, h_t^{\text{ret}})$$

Training maximizes the ELBO plus behavioral cloning on expert actions:

$$\mathcal{L} = \mathbb{E}_{q_{\phi}} \left[\sum_t \log \pi_{\psi}(a_t | z_t^{(K_t)}) - \beta \sum_t \text{KL}(q_{\phi} \| p_{\theta})_t - \lambda_c \sum_t K_t \right]$$

where the $\lambda_c \sum_t K_t$ term penalizes excess computation, incentivizing economy in deliberation length.

Learning or Inference Procedure

1. Collect offline demonstrations of target tasks.
2. Train prior p_{θ} , posterior q_{ϕ} , policy π_{ψ} , and halt predictor p_h jointly via ELBO on demonstration trajectories.
3. At inference, use the prior p_{θ} only (no access to future): sample $z_t^{(k)} \sim p_{\theta}(\cdot | z_t^{(1:k-1)}, o_t)$ and halt when $p_h(z_t^{(k)}, o_t) > u$ for $u \sim U[0, 1]$.
4. Decode the final latent state to action via $\pi_{\psi}(a_t | z_t^{(K_t)})$.
5. Fine-tune with online RL (SAC) to improve beyond demonstrations.

What the Guarantee Says

LMP does not provide formal sample-complexity bounds. The paper proves empirically that: (1) The average deliberation depth $\bar{K}$ correlates positively with task difficulty (measured by expert-labeled complexity scores, Pearson $\rho = 0.73$ across tasks). (2) Performance improves monotonically as the maximum allowed deliberation budget $K_{\max}$ increases, with diminishing returns above $K_{\max} = 8$ for the tested tasks. (3) The computation overhead of LMP versus a fixed-depth LSTM is at most $2\times$ wall-clock time, as deliberation steps are shallow and the learned halting mechanism allocates deep deliberation only to a minority of high-complexity decisions.

Experimental Findings

Task	LMP	LSTM	S4	Dreamer
Stacking (5 obj.)	71.3%	48.5%	52.1%	44.2%
Multi-step tool use	63.7%	41.2%	45.6%	38.9%
Long-horizon P&P	78.4%	56.8%	61.3%	52.7%
Deformable manip.	55.1%	32.4%	37.8%	28.6%

Success rates (%) averaged over 5 seeds and 200 evaluation episodes. LMP exceeds the next-best baseline (S4) by 16–26% across all tasks, with the largest gains on deformable manipulation where unexpected contact events require the most reactive deliberation.

Ablations and Interpretation

Fixed vs. adaptive K : Replacing the learned halting mechanism with fixed $K = 4$ (the mean deliberation depth under adaptive halting) reduces performance by 11% on average, confirming that the *alignment* of depth with difficulty, not just depth on average, drives the gains.

ELBO vs. BC-only: Removing the ELBO training signal and using behavioral cloning alone (no latent structure) reduces success rates to near-LSTM levels, confirming the structured latent representation is essential.

Computation cost: The deliberation overhead is $1.4\text{--}1.9\times$ LSTM wall-clock time (task-dependent). The halting mechanism allocates $K \geq 6$ steps to only 12% of decisions, keeping average cost low.

Transferability: LMP policies pre-trained on one task set transfer to related tasks with 40% less fine-tuning data than LSTM baselines, suggesting the latent memory structure captures more transferable task representations.

Reference

Chuning Zhu, Eva Xu, Jose Barreiros, Krishnan Srinivasan, Paarth Shah, Abhishek Gupta. **Latent Memory Palace: Reasoning for Control as Autoregressive Variational Inference.** arXiv:2607.08724, July 2026. <https://arxiv.org/abs/2607.08724>